Practical QML in the NISQ Era: From Static Encoding to Automated Hybrid Pipelines

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Problem Formulation: The NISQ Bottleneck

The Hardware Challenge

NISQ devices lack Quantum RAM. Native data processing is severely constrained by limited qubit budgets and high gate error rates.

Step 1: Baseline

- **Encode:** Implemented a linear-time $\log_2 N$ encoding technique for classical-to-quantum embedding.
- **Execute:** Successfully ran a simplified Quantum Kernel SVM (sqKSVM).
- **Limit:** While feasible, global encoding restricts this baseline to small, static tabular datasets.

sqKSVM Mathematics

1. **Data Encoding:** Map $\vec{x} \in \mathbb{R}^N$ to $n = \log_2 N$ qubits:

$$|\psi_{\vec{x}}\rangle = U(\vec{x})|0\rangle^{\otimes n}$$

2. **Quantum Kernel:** Fidelity (overlap squared):

$$K(\vec{x}_i, \vec{x}_j) = |\langle \psi_{\vec{x}_i} | \psi_{\vec{x}_j} \rangle|^2$$

3. **Decision Rule:** Analytical classification output:

$$\tilde{y} = sgn\left(\sum_{i=1}^{M} y_i \alpha_i{}^* K(\vec{x}_i, \vec{x})\right)$$

Problem Formulation: The High-Dimensional Data Problem

The Scaling Limitation

Real-world signals, such as continuous audio and speech, are massive and temporally complex. Attempting global quantum encoding of raw audio waveforms instantly exceeds current NISQ qubit limits.

Our Engineering Objective

- **The Question:** How do we scale quantum feature extraction to high-dimensional, temporal data without requiring deeper, error-prone quantum circuits?
- **The Pivot:** Moving from static clinical data to dynamic Automatic Speech Recognition (ASR).

Technique Explanation: Decentralizing Extraction (Step 2)

Quantum Convolution (QCNN)

- **Architecture:** Localized "Quanvolutional" filters replacing global data encoding.
- **Execution:** A parameterized circuit acts as a sliding window across the 2D-spectrogram.

Quantum Sieve Mathematics

1. **State Encoding:** Map 2×2 patch to 4 qubits:

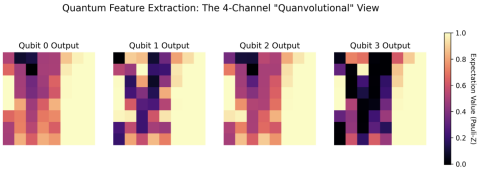
$$|\psi_{in}\rangle = e(u_x)$$

2. **Entanglement:** Apply parameterized gates:

$$|\psi_{out}\rangle = U(\vec{\theta})|\psi_{in}\rangle$$

3. **Measurement:** Pauli-Z expectation value:

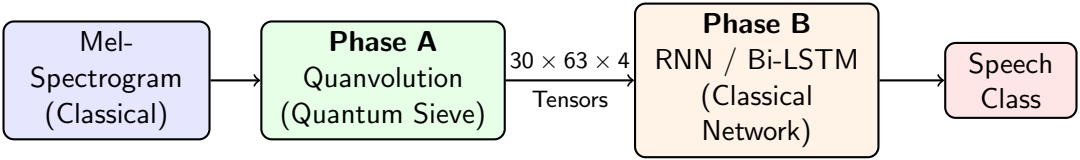
$$f_i = \langle\psi_{out}|\sigma_z|\psi_{out}\rangle$$



(Fig 1: Quanvolution Sliding Window)

Visualizes the 4 distinct spatial feature maps (Pauli-Z expectation values) extracted by the 4 qubits.

Technique Explanation: The Hybrid CQ Architecture



Architectural Advantage

The QPU acts purely as a non-linear feature extractor. The extracted tensors are collapsed classically and fed into an RNN. This perfectly respects NISQ hardware constraints by maintaining shallow circuit depth.

Phase B: Sequence Mathematics

1. RNN Update Rule:
Processes the flattened quantum sequence (f_t) to update hidden states (h_t):

$$h_t = \text{LSTM}(f_t, h_{t-1})$$

2. Classification:
Softmax attention over the hidden states yields the 10-class speech prediction.

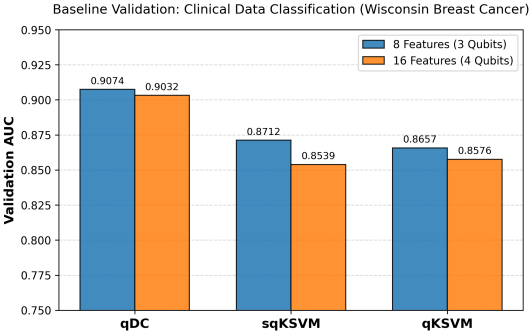
Results: Step 1 Baseline Classification

Clinical sqKSVM Metrics

- **Dataset:** Tabular clinical records.
- **Encoding:** Linear-time $\log_2 N$ state preparation.
- **Outcome:** Successfully classified data points utilizing a strict qubit budget.

Validation

Proves the feasibility of mathematical data encoding on constrained hardware.



(Fig 2: sqKSVM Classification Results)

Demonstrates successful binary classification of clinical data boundaries using a strict quantum kernel.

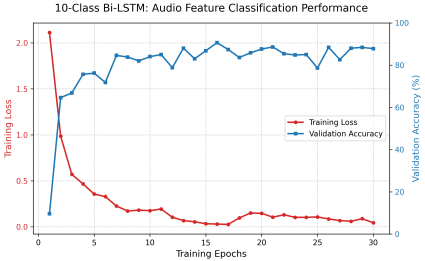
Results: Step 2 Benchmark Replication (Cloud)

10-Class Sequence Training (Keras Bi-LSTM)

Processed the full 3,000-file dataset through the Bi-LSTM sequence architecture, relying entirely on the extracted quantum spatial tensors to successfully replicate the published baseline.

Training Metric	Achieved Value
Total Parameters	180,437
Dataset Classes	10 (Full Audio)
Peak Validation Epoch	Epoch 16
Training Accuracy	99.07%
Validation Accuracy	**90.60%**

Table 1: Final empirical 10-class results.



(Fig 3: Bi-LSTM Epoch Metrics)

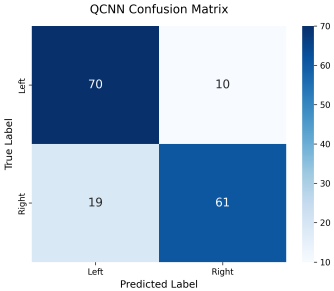
Model convergence hitting peak val. acc. at Epoch 16.

Results: Step 2 Local Hardware Validation (PyTorch)

Engineering Prototype: Local Hardware Validation

Prior to executing the computationally expensive 10-class extraction on the Linux server, the quantum spatial features were mathematically validated locally.

- **Subset Testing:** Processed a binary audio subset (Left vs. Right).
- **Architecture:** Built a custom, lightweight PyTorch CNN to ingest the 4-channel quantum tensors.
- **Outcome:** Confirmed strong feature separation, proving the quantum circuit's validity before cloud scaling.



(Fig 4: Binary Confusion Matrix)

Proves strong subset separation utilizing the custom

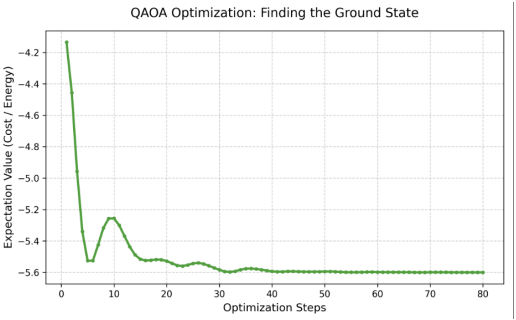
Novelty: Automating the Pipeline with QAOA (Step 3)

The Operational Friction

Manually tuning hyperparameters across a delegated CPU/QPU architecture is unscalable for production.

The QAOA Solution

- Formulated Hyperparameter Optimization as a QUBO problem.
- Deployed the **Quantum Approximate Optimization Algorithm (QAOA)** to navigate the parameter space autonomously.



(Fig 5: QAOA Optimizer Convergence)

Illustrates the QAOA solver dynamically minimizing the cost function.

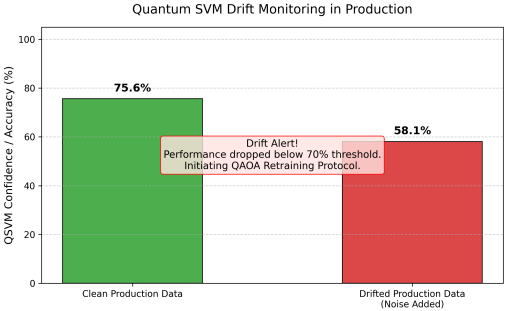
Novelty: Closing the Loop with Production Monitoring

Repurposing the Baseline

We brought back the **Quantum SVM** from Step 1, completely shifting its role. Instead of clinical classification, it now acts as a real-time boundary monitor to detect data drift in live audio streams.

Final Automated Architecture

1. **QCNN + RNN:** Acoustic transcription.
2. **QAOA:** Continuous optimization.
3. **QSVM:** Instant drift flagging.



(Fig 6: QSVM Production Drift Detection)

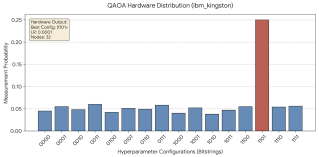
Displays the QSVM boundary successfully flagging out-of-distribution audio samples.

Hardware Validation: IBM QPU Execution

Executing on Real Quantum Hardware

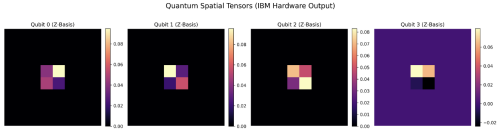
To validate the hybrid pipeline on actual NISQ devices, aggressive optimizations were deployed to survive queue times and hardware coherence limits.

QAOA Optimization (ibm_kingston)



- **Hardware Adaptation:** Reduced optimization steps to 2 and circuit depth to 1 to mitigate high decoherence rates.
- **Execution:** Successfully executed the classical-quantum feedback loop on the `ibm_kingston` device.

QCNN Spatial Tensors



- **Center-Crop Strategy:** Isolated the center 4×4 frequency patch to drop the workload from 64 to exactly 4 quantum circuits.
- **Safety Padding:** Embedded the results into an 8×8 zero matrix to maintain downstream shape compatibility.

Summary & References

Project Summary

1. **Feasibility:** Validated NISQ constraints using linear-time encoding (sqKSVM).
2. **Scalability:** Scaled to ASR using local QCNN feature extraction combined with classical RNNs.
3. **Deployability (Novelty):** Automated the entire hybrid CQ pipeline using QAOA for HPO and QSVM for drift monitoring.

Key References

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